

# *Teaching Notes 0\**

## *Constructing Cash Flows*

Pietro Veronesi  
The University of Chicago  
Booth School of Business

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# Overview

- Types Of Cash Flows
  - Debt, Equity and Firm Value
  - Free Cash Flow (FCF) → Enterprise Value
- Rules For Calculating Cash Flows.
  - Financial Cash Flows and Accounting Earnings
- Determining Free Cash Flows
  - Deriving The Formula and Alternative Expressions
  - Examples
  - Subtle Points
- Which Discount Rates?
- Other Cash Flows
- Example

# Objectives

- Understand construction of Free Cash Flows (FCF)!
  - Go over the different components of the FCF formula. Be sure to have them right!
- Apply different rules for constructing cash flows

# Big Picture: How to Value a Project/Firm?

- Look up the price of a comparable project:
  - Apply Law of One Price
  - Using comparables/multiples (More later in the course)
- Calculate Net Present Value:
  - Estimate the expected/projected cash-flows.
  - Estimate the appropriate discount rate for each cash flow.
  - Calculate NPV.
- Now lets first concentrate on **calculating cash flows**
  - Will build up understanding on discount rates

# Three Types of Cash Flows

# Three Types Of Cash Flows

- There are three types of value we care about in Corporate Finance and Valuation:
  - Market Value of Debt,  $V_D$
  - Market Value of Equity,  $V_E$
  - Total Market Value or Enterprise Value,  $V = V_D + V_E$
- Depending on the value you are interested in, you will construct a different cash flow.
  - Free Cash Flow (FCF) → Enterprise Value.
  - *Capital Cash Flow (CCF) → Enterprise Value.*
  - *Equity Cash Flow (ECF) → Equity Value.*
- Today we will concentrate on the simplest (and most commonly used) cash flow: **Free Cash Flow.**

# Rules for Calculating Cash Flows

# Financial Cash Flows And Accounting Earnings

- The value of a firm stems from the cash flow that it generates.
  - Calculating cash flows from Income Statements and Balance Sheets is the most important part of a valuation.
- It is important to distinguish between **income** and **cash flow**.
  - The difference relies in the accrual accounting methods.
- Accrual accounting is **“transaction based”**.
  - Records information on financial statements when transactions occur, not necessarily when a party is paid (receives cash) for the transaction.
  - **Example:** A sale of an engine is recorded when it occurs, even though the customer gets trade credit and seller will not receive the cash for several days.
- **Bottom Line:** We need to make various adjustments to accrual based measures of income to extract the relevant information.



# Simple Rules for Calculating Cash Flows

- **Rule 1:** Use **only** cash flows attributable to the project.
  - Cash flows should be **incremental** to the project.
  - **Forget Sunk Cost:** bygones are bygones.
- **Rule 2:** Use **all** cash flows attributable to the project.
  - **Include opportunity costs** of using existing equipment, facilities as costs.
    - Example: Land which could be sold or rented.
  - Effects on **working capital**.
    - If project requires follow-on CAPX or additional working capital take these into account.

# Simple Rules for Calculating Cash Flows (cntd)

- **Rule 3:** Use **cash flows**, not accounting earnings.
  - Cash Flows: Show money in and out at any point in time.
  - Accounting: Show when money is earned.
    - Useful to make a representation of the business.
    - Not useful for valuation purposes.
  - **Depreciation** affects cash flows only through its effect on taxes.
- **Rule 4:** Be consistent in the treatment of **inflation**.
  - Discount nominal cash flows at nominal discount rates.
- **Bottom Line:** **Count** all **incremental, after-tax** cash flows allowing for reasonable **inflation**.

# Determining Free Cash Flows

# Free Cash Flow Formula

- The FCF for a single period, typically one year, is given by:

$$FCF_t = (1 - t) \times EBIT_t + Depreciation_t - CAPX_t - \Delta NWC_t + Other$$

- Note:

- EBIT = Earnings Before Interest and Taxes
- $t$  = Tax Rate, generally the “marginal corporate tax rate”.
- CAPX = Capital Expenditures
- $\Delta NWC$  = Change in Net Working Capital.
- The “ $t$ ” subscript denotes the year or month. Likewise, “ $t-1$ ” denotes the previous year or month.

# Free Cash Flow Formula (cont.)

- The FCF is the amount of funds available to all investors in a firm after the firm pays taxes and meets investment needs.
  - We start with EBIT because this is what is available to be paid to the owners – regardless of the existing debt/equity mix .
  - We **do not** want tax consequences of capital structure to matter.
- The expected after-tax cash flows of an **all-equity** firm.
  - It is a **hypothetical** cash flow.
  - It is the **actual** cash flow for a 100% equity financed firm.
- These cash flows ignore the tax savings the firm gets from debt financing (the deductibility of interest expense).
  - **Will account for this effect in different ways subsequently**

# Free Cash Flow Formula (cont.)

- **Two Steps towards figuring out firm value:**
  - **Step 1:** Estimating the Free Cash Flows.
  - Step 2: Account for the effect of financing on value (later)

# Determining Free Cash Flows

- Let's first consider the cash flows for an all-equity financed project so there are no interest payments to worry about.
  - Will ignore considerations related to taxation of interest on debt.
- The project's FCF in period  $t$  are given by:

$$FCF_t = (\text{Cash Inflows})_t - (\text{Cash Outflows})_t$$

$$(1) FCF_t = (\text{Cash Flow From Sales})_t - (\text{Cash Outlay For Goods})_t - (\text{Operating Expenses})_t - (\text{CAPX})_t - (\text{Taxes Paid})_t$$

- We need to calculate each of these items for a single year for a simple business.
  - Let us fill the **five** items of the above equation using an income statement and balance sheet.

# Determining Free Cash Flows (cont.)

## Simplified Accounting Statement Required To Compute FCF

| Income Statement (t)                                              |  | Balance Sheet (t)                |                               |
|-------------------------------------------------------------------|--|----------------------------------|-------------------------------|
|                                                                   |  | Assets                           | Liabilities                   |
| Sales <sub>t</sub>                                                |  | Accounts Receivable <sub>t</sub> | Accounts Payable <sub>t</sub> |
| - Cost of Goods Sold <sub>t</sub> (COGS <sub>t</sub> )            |  | Inventories <sub>t</sub>         |                               |
| - Operating Expense <sub>t</sub>                                  |  | PPE <sub>t</sub>                 |                               |
| = Operating Profit (EBITDA)                                       |  |                                  |                               |
| - Depreciation <sub>t</sub>                                       |  |                                  |                               |
| = EBIT <sub>t</sub> ( = EBT <sub>t</sub> if Interest Expense = 0) |  |                                  |                               |
| - Taxes <sub>t</sub>                                              |  |                                  |                               |
| = Net Income <sub>t</sub>                                         |  |                                  |                               |

- **Note:** In this particular case of an all-equity financed firm,  $EBIT_t = EBT_t$  since Interest Expense is **zero**!



# Determining Free Cash Flows (cont.)

- **Cash Flow From Sales:** Sales may take time to get paid. Sales not paid immediately are recorded as account receivables (AR).

$$(2) \text{ Cash Flow From Sales}_t = \text{Sales}_t - (\text{AR}_t - \text{AR}_{t-1})$$

- A positive change in account receivables is a **use** of funds!
- **Cash Outlay For Goods:** Not the same as the COGS because:
  - Those goods may not result in a cash transaction.
  - The firm may purchase more or less than it register in sales, holding the remainder in inventory.

$$\text{Cash Outlay of Goods}_t = \text{Purchases}_t - (\text{AP}_t - \text{AP}_{t-1})$$

- A positive change in account payable (AP) is a **source** of funds!

# Determining Free Cash Flows (cont.)

- **Cash Outlay For Goods (cont'd):**

- Using Account Payables to Finance is also known as “Trade Credit”.
- Substituting the expression of the Cash Outlays<sub>t</sub> in the accounting identity  $\text{Purchases}_t = \text{COGS}_t + (\text{Inv}_t - \text{Inv}_{t-1})$  we get:

$$(3) \text{ Cash Outlay of Goods}_t = \text{COGS}_t + (\text{Inv}_t - \text{Inv}_{t-1}) - (\text{AP}_t - \text{AP}_{t-1})$$

- A positive change in Inventories is a **use** of funds!
- **Bottom Line:** There are timing differences between the accounting measures “Sales”/“COGS” and cash flows.
  - Changes in net working capital ( $\Delta\text{NWC}$ ) account for difference.
    - $\Delta\text{NWC}$  are **cash flows** that must be accounted for.

$$\Delta\text{Net Working Capital}_t = \Delta\text{Inv}_t + \Delta\text{AR}_t - \Delta\text{AP}_t$$

# Determining Free Cash Flows (cont.)

- **Operating Expenses:** Include income statement lines such as selling, general, and administrative expenses (SG&A), as well as research and development expense (R&D).
- **Capital Expenditures:** CAPX do **not** appear directly in the Income Statement because they are capitalized expense.
  - They show up via depreciation on the income statement and as increases in Property, Plant and Equipment (PPE) on the Balance Sheet. The link between these terms is given by
$$PPE_t = PPE_{t-1} + CAPX_t - Depreciation_t$$
  - However, they are a cash outlay and they are part of the FCF.
- **Question:** What about Depreciation?

# Determining Free Cash Flows (cont.)

- **Remark:** We do **not** include accounting depreciation as an expense since it is not a cash flow.
  - However accounting depreciation does affect cash flow because it **reduces** the company's tax bill.
- **Taxes Paid:** Taxes are paid on income statement earnings, EBIT.
  - This is only true if the firm is an all equity-financed firm!
    - Taxes are paid on EBT and not EBIT, but in this particular case:  
EBIT = EBT (Earnings Before Taxes).
  - **However**, even if the firm has debt financing, we will calculate taxes **as if** the firm were all equity (on EBIT) and then add in separately the tax benefits of financing.

$$(4) \text{ Taxes Paid}_t = t \times (\text{Sales}_t - \text{COGS}_t - \text{Operating Expense}_t - \text{Depreciation}_t)$$

# Determining Free Cash Flows (cont.)

- Substituting equations (2), (3) and (4) in (1):

$$FCF_t = (\text{Cash Flow From Sales})_t - (\text{Cash Outlay For Goods})_t - (\text{Operating Expenses})_t - (\text{CAPX})_t - (\text{Taxes Paid})_t$$

$$FCF_t = \text{Sales}_t - (\text{AR}_t - \text{AR}_{t-1}) - \text{COGS}_t - (\text{Inv}_t - \text{Inv}_{t-1}) + (\text{AP}_t - \text{AP}_{t-1}) - (\text{Operating Expenses})_t - (\text{CAPX})_t - t \times (\text{Sales}_t - \text{COGS}_t - \text{Operating Expense}_t - \text{Depreciation}_t)$$

$$FCF_t = (1 - t) \times (\text{Sales}_t - \text{COGS}_t - \text{Operating Expenses}_t - \text{Depreciation}_t) - \text{CAPX}_t + \text{Depreciation}_t - (\text{AR}_t - \text{AR}_{t-1}) - (\text{Inv}_t - \text{Inv}_{t-1}) + (\text{AP}_t - \text{AP}_{t-1})$$

$$FCF_t = (1 - t) \times (\text{Sales}_t - \text{COGS}_t - \text{Operating Expenses}_t - \text{Depreciation}_t) - \text{CAPX}_t + \text{Depreciation}_t - \Delta \text{AR}_t - \Delta \text{Inv}_t + \Delta \text{AP}_t$$



$$FCF_t = (1 - t) \times \text{EBIT}_t - \text{CAPX}_t + \text{Depreciation}_t - \Delta \text{NWC}_t$$

- Note:**  $\text{EBIT}_t = (\text{Sales}_t - \text{COGS}_t - \text{Operating Expenses}_t - \text{Depreciation}_t)$

# Determining Free Cash Flows (cont.)

## Final Remarks

- **Remark 1:** The FCF formula is a convenient representation that allows one to compute the FCF in a few simple steps.
  - It requires a complete forecast of the income statement.
  - It requires only information on changes in non-financing related assets and liabilities from the balance sheet.
- **Remark 2:** Do not forget changes in working capital.
  - Changes in working capital are cash flows (similar to capital expenditures) that must be accounted for.
- **Remark 3:** Accounting depreciation affects cash flows only by reducing the company's tax bill.
- **Remark 4:** FCFs used to compute **Enterprise Value**.

# Free Cash Flows: Equivalent Expressions

- You may also encounter the FCF formula in the following **equivalent** forms:

- Using:

- $EBIT_t = EBITDA_t - \text{Depreciation}_t$

$$FCF_t = (1 - t) \times EBITDA_t + t \times \text{Depreciation}_t - CAPX_t - \Delta NWC_t$$

- Using:

- $CAPX_t + \Delta NWC_t - \text{Dep}_t = \Delta \text{Net Assets}_t$

- $\text{Net Assets}_t = \text{Net PPE}_t + NWC_t$

$$FCF_t = (1 - t) \times EBIT_t - \Delta \text{Net Assets}_t$$

- Using:

- $NOPAT = EBIT(1 - t)$ ; [NOPAT<sub>t</sub> = Net Operating Profit After Tax]

- It differs from “Net Income” in that interest expenses are not considered in NOPAT.

$$FCF_t = NOPAT_t + \text{Depreciation}_t - CAPX_t - \Delta NWC_t$$

## Example I: Depreciation/Accounting Earnings

- **Point:** Depreciation is treated **differently** for cash-flow purposes than it is for accounting purposes.
- **Example:** A machine is purchased for \$1,000,000 with a life of 10 years. Each year it produces revenues of \$300,000 and has operating expenses of \$100,000. Assume that the machine will be depreciated over 10 years using straight-line depreciation. The corporate tax rate is 40%.
  - **Question:** What are your annual accounting earnings and your annual FCF?



# Example I: Depreciation/Accounting Earnings

- The annual earnings and FCF (in \$,000) are therefore:

| Date | Accounting Earnings Before Tax                 | Accounting Earnings After Tax           | Free Cash Flow                             |
|------|------------------------------------------------|-----------------------------------------|--------------------------------------------|
|      | = Revenues - Operating Expenses - Depreciation | = Accounting Earnings Before Tax *(1-t) | =EBIT(1-t) + Dep - CAPX - ΔNWC             |
| 0    | 0                                              | 0                                       | -1,000                                     |
| 1    | = 300 - 100 - 100 = <b>100</b>                 | = (1 - 0.4)*100 = <b>60</b>             | =(1 - 0.4)(300-100-100) + 100 = <b>160</b> |
| 2    | 100                                            | 60                                      | 160                                        |
| 3    | 100                                            | 60                                      | 160                                        |
| 4    | 100                                            | 60                                      | 160                                        |
| 5    | 100                                            | 60                                      | 160                                        |
| 6    | 100                                            | 60                                      | 160                                        |
| 7    | 100                                            | 60                                      | 160                                        |
| 8    | 100                                            | 60                                      | 160                                        |
| 9    | 100                                            | 60                                      | 160                                        |
| 10   | 100                                            | 60                                      | 160                                        |

## Example II: Computing FCF

|                      | T=0  | T=1  |
|----------------------|------|------|
| Sales                | 1000 | 1200 |
| Cost of Goods Sold   | 700  | 850  |
| Depreciation         | 30   | 35   |
| Interest Expense     | 40   | 50   |
| Taxes @ 38%          | 80   | 90   |
| Profit After Taxes   | 150  | 175  |
| Capital Expenditures | 40   | 40   |
| Accounts Receivable  | 50   | 60   |
| Inventories          | 50   | 60   |
| Accounts Payable     | 20   | 25   |

- In time period T=1:
  - $FCF = EBIT \times (1 - t) + \text{Depreciation} - \text{CAPX} - \Delta NWC + \text{Other}$ 
    - $EBIT = \text{Sales} - \text{COGS} - \text{Depreciation} = \$1,200 - \$850 - \$35 = \$315$
    - $t = 38\%$
    - $\Delta NWC = (\$60 + \$60 - \$25) - (\$50 + \$50 - \$20) = \$15$
  - $FCF = \$315 \times (1 - 0.38) + \$35 - \$40 - \$15 = \$175.3$

# Free Cash Flow: Beware!

- **Note:**
  - We ignored interest payments.
  - We computed taxes on EBIT.
- Do **not** take the effect of financing (e.g., interest) into account at this stage. Why?
- Remember our plan:
  - First, determine the expected cash-flows as if the project were 100% equity financed.
  - Later, we will adjust for financing.
- **Remark:** If you count financing costs in cash-flow, you count them **twice!**
  - Since we will take care of the financing costs in the discount rate!

# Additional Points

- **Accrued Taxes/Deferred Taxes:**

- When taxes are incurred on the income statement but not immediately paid, they are recorded as increases in deferred taxes.
  - They are treated separately as sources of cash in the free cash flow formula.
- Should be added back since it is a non-cash expense. This is tax that we owe, but do not presently have to pay.

$$FCF_t = (1 - t) \times EBIT_t - CAPX_t + \Delta \text{Deferred Taxes}_t + \text{Depreciation}_t - \Delta NWC_t$$

# Additional Points

- **Terminal Value**

- Idea is that at horizon's end, a project may still have some value.

- Two Methods to account for this value

- If project is liquidated or sold at the end of the horizon, we calculate its salvage value (“**Liquidation Method**”)
- If the project will continue indefinitely, we calculate its continuation value (“**Perpetuity Method**”)

# Additional Points

- **Terminal Value: Method 1 (“Salvage Value” Method)**
  - Appropriate method if machine or project is going to be sold off at the end of the horizon
  - Best for slow growing companies
- Start with estimate of sales price of machine in the year “after” the final forecasted cash flow
- **Consider tax implications**
  - If selling price < book value, firm realizes a tax shield on capital loss
  - If selling price > book value, firm pays tax on capital gain
  - If selling price = book value → no tax liability

# Additional Points

- **Terminal Value: Method 2 (“Continuation Value” Method)**
  - Appropriate method if investment continues beyond the horizon where components of cash flow easily calculated
  - Best to use if cash flows have settled into a pattern of steady growth at the end of the forecast horizon
- Take the cash flows at time  $T$ , calculate (or assume) a growth rate  $g$  for them, and apply the growing perpetuity formula

# Additional Points

- **Terminal Value: Method 2 (“Continuation Value” Method)**
- Suppose that  $r$  is 14% and we have the following cash flows

|                                     | 0     | 1    | 2    | 3    | 4    | 5       |
|-------------------------------------|-------|------|------|------|------|---------|
| Cash Flows Excluding Terminal Value | -3500 | 1000 | 1150 | 1158 | 1212 | 1275    |
| Continuation Value                  |       |      |      |      |      | CV      |
| Total Cash Flows                    | -3500 | 1000 | 1150 | 1158 | 1212 | 1275+CV |

- Growth rate of cash flows?
  - Roughly 5%
- What is continuation value (CV) here?

$$\frac{CF(1 + g)}{(r - g)} = \frac{1275(1 + 0.05)}{(0.14 - 0.05)} = 14,875$$



Which Discount Rate?

# Which Discount Rate?

- We discount the FCF using the project's cost of capital,  $r$ .
  - “Opportunity cost of capital” for capital providers
- If the project is **all-equity financed**:
  - Use  $r = r_E$  where  $r_E$  is the expected return of equity holders.
- If the project is **all-debt financed**:
  - Use the same cash flow formulas as above.
  - Use  $r = r_D$  where  $r_D$  is the expected return of debt holders
    - We would have to adjust this for “debt tax shields” (later)
- **Later:** What if project is partly equity and partly debt financed?

# Other Cash Flows

# Capital Cash Flows

- An alternative to using the same cash flows but at a tax-adjusted cost of capital is to use adjusted cash flows and no tax-adjustments to the cost of capital.
  - This cash flow is known as **Capital Cash Flow** (CCF).
  - Some practitioners call it **actual** cash flow since taxes are computed on EBT ( $t \times EBT$ ) and not EBIT ( $t \times EBIT$ ) as in FCF.
- In FCF:
  - Construct the cash-flow of 100% equity financed project even if it is not 100% equity financed!
  - Adjust for financing effects in the cost of capital.
- In CCF:
  - Adjust the cash flow taking into account the financing effect.
  - Do not adjust the cost of capital for financing effect.

# Free Cash Flow vs. Capital Cash Flow

## EBIT Version

- **$FCF_t$**  =  $EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t$
- **$CCF_t$**  =  $FCF_t + t \times \text{Interest Expense}_t$ 
  - $CCF_t = EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t + t \times Int_t$
  - $CCF_t = EBIT_t - t \times EBIT_t + t \times Int_t + Dep_t - CAPX_t - \Delta NWC_t$
  - **$CCF_t = EBIT_t - t \times EBT_t + Dep_t - CAPX_t - \Delta NWC_t$**

# Equity Cash Flows

- **Equity Cash Flows** (ECF) measure the cash flow available to stockholders after payments to debt holders are deducted from operating cash flows.
- Payments to debt holders are sometimes called **Debt Cash Flows** and they include interest and principal payments.

$$ECF_t = \text{Capital Cash Flow}_t - \text{Debt Cash Flow}_t$$

- ECFs calculated by subtracting taxes, interest and debt repayments from operating cash flows and adding debt additions.
- **Remark:** Discount rate used in the Equity Cash Flow methodology is the return on equity  $r_E$  and the value we obtain is **Equity Value**.

# Free Cash Flow vs Equity Cash Flow

## EBIT Version

- **FCF<sub>t</sub>** =  $EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t$
- **CCF<sub>t</sub>** =  $EBIT_t - t \times EBT_t + Dep_t - CAPX_t - \Delta NWC_t$
- **ECF<sub>t</sub>** =  $EBIT_t - t \times EBT_t + Dep_t - CAPX_t - \Delta NWC_t - [Debt Payments_t - Debt Issues_t]$

Example



# Turbo Widgets (TW)

- XYZ, a profitable widget producer (\$100M annual after-tax profit) contemplates introducing new Turbo Widgets (TWs), developed in its labs at an R&D cost of \$1M over the past 3 years.
- New plant to produce TW would:
  - Cost \$20M today.
  - Last 10 years with salvage value of \$5M.
  - Be depreciated to \$0 over 5 years using straight-line.
- TWs need painting: Use 40% of the capacity of a painting machine:
  - Currently owned and used by XYZ at 30% capacity.
  - With maintenance costs of \$100,000 (regardless of used capacity)

# TW: Other Parameters

- Annual:
  - Operating Costs: \$400,000
  - Operating Income Generated (\*): \$42M
    - You may want to assume that operating income = Sales Revenue.
  - Sales of Regular Widgets would decrease by \$2M
- Working capital (WC): \$2M needed over the life of the project.
- Corporate tax rate 36%.

## TW Example: Use Incremental Cash Flows

- Compare firm value with and without the project:

$$V(\text{Project}) = V(\text{Firm w/project}) - V(\text{Firm w/o project})$$

- Use **only** cash flows (in and out) **attributable** to the project.
- In particular (among others):
  - Sunk costs should be ignored.
  - Include opportunity costs.

## TW Example: Use Incremental Cash Flows

- **Ignore** the \$100M after-tax profit and focus on incremental cash-flows.
- R&D cost of \$1M over the past three years: Sunk cost. **Ignore it!**
- The plant's \$20M cost: It's a CAPX. **Count it!**
- Machine's \$100K maintenance cost: Not incremental. **Ignore it!**
  - Incurred with or without TW production.

# TW Example: All Cash Flows

- Sales of regular widgets decrease by \$2M due to cannibalization.
  - Would not occur without TW production.

| Year              | 0  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-------------------|----|---|---|---|---|---|---|---|---|---|----|
| CAPX              | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  |
| RW Sales decrease | 0  | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2  |

# TW Example: Use After-Tax Cash Flows

- These are what you have left after paying capital suppliers.
- Make sure to count the benefits of expensing, depreciation, etc.
- CAPX and Depreciation:
  - Tax treatment of CAPX depends on depreciation
  - Each year, depreciation is the fraction of CAPX that can be subtracted from that year's taxable income

# TW Example: Depreciation and CAPX

- Depreciation:
  - Straight line depreciation: Flat annual depreciation.
  - Accelerated depreciation: Decreasing.
- \$20M CAPX is depreciated linearly over 5 years, down to zero.

$$\text{Depreciation}_t = (\$20 - \$0) / 5 = \$4\text{M}$$

- Salvage value \$5M is fully taxable since book value is zero.
  - Get tax credit if  **$t^*(\text{PPE-Salvage Value}) > 0!$**

| Year          | 0  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---------------|----|---|---|---|---|---|---|---|---|---|----|
| CAPX          | 20 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  |
| Depreciation  | 0  | 4 | 4 | 4 | 4 | 4 | 0 | 0 | 0 | 0 | 0  |
| Salvage Value | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5  |

# TW Example: Putting The Pieces Together

| Year                    | 0            | 1           | 2           | 3           | 4           | 5           | 6           | 7           | 8           | 9           | 10          |
|-------------------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CAPX                    | <b>20.0</b>  | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| Income                  | -            | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        |
| RW Sales decrease       | -            | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         |
| Incremental income      | -            | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        |
| Incremental cost        | -            | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         |
| Salvage value           | -            | -           | -           | -           | -           | -           | -           | -           | -           | -           | 5.0         |
| Incremental profit      | -            | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>44.6</b> |
| Depreciation            | -            | 4.0         | 4.0         | 4.0         | 4.0         | 4.0         | -           | -           | -           | -           | -           |
| EBIT                    | -            | 35.6        | 35.6        | 35.6        | 35.6        | 35.6        | 39.6        | 39.6        | 39.6        | 39.6        | 44.6        |
| Incremental taxes (36%) | -            | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>14.3</b> | <b>14.3</b> | <b>14.3</b> | <b>14.3</b> | <b>16.1</b> |
| Total                   | <b>-20.0</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>28.5</b> |

- Remark: We are **implicitly** assuming that the project is 100% equity financed!



# Just To Make Sure Things Are Clear...

- We could have computed the CF in year 1 following the FCF formula:

$$FCF_1 = EBIT_1 (1 - t) + Dep_1 - CAPX_1$$

$$FCF_1 = \$35.6 (1 - 0.36) + \$4 - \$0 = \$26.8$$

- **Question:** Are we done yet? **No!** Why?

## TW Example: Changes in Net Working Capital

- Many projects need some capital to be tied up (working capital) which constitutes an opportunity cost.
- We need the Change in Working Capital implied by the project.

$$\text{Net Working Capital}_t = \text{Inventory}_t + \text{AR}_t - \text{AP}_t$$

$$\Delta \text{Net Working Capital}_t = \Delta \text{Inventory}_t + \Delta \text{AR}_t - \Delta \text{AP}_t$$

| Year                    | 0            | 1           | 2           | 3           | 4           | 5           | 6           | 7           | 8           | 9           | 10          |
|-------------------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CAPX                    | 20.0         | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| Incremental profit      | -            | 39.6        | 39.6        | 39.6        | 39.6        | 39.6        | 39.6        | 39.6        | 39.6        | 39.6        | 44.6        |
| Incremental taxes (36%) | -            | 12.8        | 12.8        | 12.8        | 12.8        | 12.8        | 14.3        | 14.3        | 14.3        | 14.3        | 16.1        |
| NWC                     | 2.0          | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | -           |
| Change in NWC           | 2.0          | -           | -           | -           | -           | -           | -           | -           | -           | -           | -2.0        |
| <b>Total</b>            | <b>-22.0</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>30.5</b> |

# In Summary...

- The FCF are summarized as follows:

| Year                       | 0             | 1           | 2           | 3           | 4           | 5           | 6           | 7           | 8           | 9           | 10          |
|----------------------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CAPX                       | <b>20.0</b>   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| Income                     | -             | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        | 42.0        |
| RW Sales Decrease          | -             | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         |
| Incremental Income         | -             | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        | 40.0        |
| Incremental Cost           | -             | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         | 0.4         |
| Salvage Value              | -             | -           | -           | -           | -           | -           | -           | -           | -           | -           | 5.0         |
| Incremental Profit         | -             | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>39.6</b> | <b>44.6</b> |
| Depreciation               | -             | 4.0         | 4.0         | 4.0         | 4.0         | 4.0         | -           | -           | -           | -           | -           |
| EBIT                       | -             | 35.6        | 35.6        | 35.6        | 35.6        | 35.6        | 39.6        | 39.6        | 39.6        | 39.6        | 44.6        |
| Incremental Taxes (36%)    | -             | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>12.8</b> | <b>14.3</b> | <b>14.3</b> | <b>14.3</b> | <b>14.3</b> | <b>16.1</b> |
| EBIT(1-t) + Dep - CAPX     | <b>(20.0)</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>28.5</b> |
| NWC                        | 2.0           | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 2.0         | 0.0         |
| <b>Minus</b> Change In NWC | 2.0           | -           | -           | -           | -           | -           | -           | -           | -           | -           | (2.0)       |
| <b>FCF</b>                 | <b>(22.0)</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>26.8</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>25.3</b> | <b>30.5</b> |

# Takeaways

- In constructing any type of cash flow there are certain rules you should follow:
  - Use all incremental cash flows attributable to the project.
  - Use cash flows, not accounting earnings.
  - Be consistent in the treatment of inflation.
- Getting the correct free cash flow estimates will usually have a much **larger** impact on the final value than obtaining the correct discount rate.
  - FCF is a **hypothetical** cash flow of an all equity-firm!
- Points to Remember
  - Do not forget  $\Delta NWC$  since they are a cash outlay, similar to capital expenditures.
  - Accounting depreciation affects CFs only by reducing the tax bill.